

LabTasksWeek3

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Course: OOP

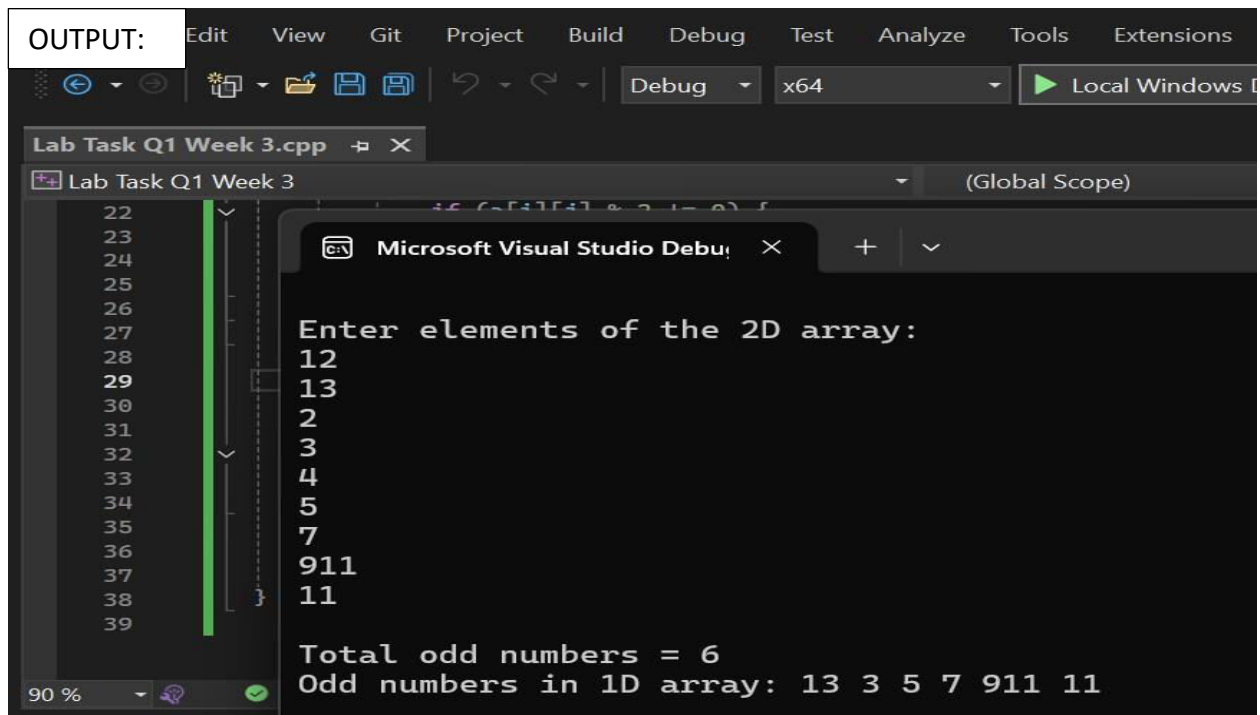
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1:

WRITE A C++ PROGRAM THAT ASKS THE USER TO ENTER THE NUMBER OF ROWS AND COLUMNS FOR A 2D INTEGER ARRAY AND THEN INPUTS ALL THE ELEMENTS OF THE MATRIX FROM THE USER. AFTER STORING THE VALUES, THE PROGRAM SHOULD CHECK EACH ELEMENT OF THE 2D ARRAY AND DETERMINE WHETHER IT IS AN ODD NUMBER OR NOT. THE PROGRAM SHOULD COUNT HOW MANY ODD NUMBERS ARE PRESENT IN THE MATRIX AND DISPLAY THE TOTAL ODD NUMBER AT THE END.

```
#include<iostream>using namespace std; int main() {    int a[3][3];    cout<<"\nEnter elements of the 2D array:\n";    for(int i = 0; i < 3; i++) {        for(int j = 0; j < 3; j++){            cin>>a[i][j];        }    }    int odd[10];    int oddCount = 0;    for (int i = 0; i < 3; i++) {        for(int j = 0; j < 3; j++){            if(a[i][j] % 2 != 0){                odd[oddCount] = a[i][j];                oddCount++;            }        }    }    cout<<"\nTotal odd numbers = "<< oddCount << endl;    cout<<"Odd numbers:";    for(int i = 0; i < oddCount; i++){        cout<< odd[i] << " ";    }    cout<< endl;    return 0;}
```

OUTPUT:



```
22  
23  
24  
25  
26  
27  
28  
29  
30  
31  
32  
33  
34  
35  
36  
37  
38  
39
```

Microsoft Visual Studio Debug Console

Enter elements of the 2D array:
12
13
2
3
4
5
7
911
11

Total odd numbers = 6
Odd numbers in 1D array: 13 3 5 7 911 11

TASK2:**WRITE A C++ PROGRAM THAT ASKS THE USER TO ENTER THE NUMBER OF ROWS AND COLUMNS FOR A 2D**

INTEGER ARRAY AND THEN INPUTS ALL THE ELEMENTS OF THE MATRIX FROM THE USER. AFTER STORING THE VALUES, THE PROGRAM SHOULD CHECK EACH ELEMENT OF THE 2D ARRAY AND DETERMINE WHETHER IT IS A PRIME NUMBER OR NOT. THE PROGRAM SHOULD COUNT HOW MANY PRIME NUMBERS ARE PRESENT IN THE MATRIX AND DISPLAY THE TOTAL PRIME.

```
#include<iostream>u
sing namespace std;
int main() {
    int rows, cols;
    cout<<"Enter number of rows:";
    cin>>rows;
    cout<<"Enter number of columns:";
    cin>>cols;
    int matrix[5][5];
    cout<<"\nEnter elements of the matrix:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cin>>matrix[i][j];
        }
    }
    int primeCount = 0;
    int primeNumbers[25];
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            int num = matrix[i][j];
            int divisors = 0;
            for (int k = 1; k <= num; k++) {
                if (num % k == 0) {
                    divisors++;
                }
            }
            if (divisors == 2) {
                primeNumbers[primeCount] = num;
                primeCount++;
            }
        }
    }
    cout<<"\nTotal prime numbers in the matrix="<<primeCount<<endl;

    if (primeCount > 0) {
        cout<<"Prime numbers are:";
        for (int i = 0; i < primeCount; i++) {
            cout<<primeNumbers[i]<<" ";
        }
        cout<<endl;
    }
    else {
        cout<<"No prime numbers found in the matrix."<<endl;
    }
    return 0;
}
```

OUTPUT:

```
Microsoft Visual Studio Debug Console
Enter elements of the matrix:
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25

Total prime numbers in the matrix = 9
Prime numbers are: 2 3 5 7 11 13 17 19 23
```

TASK3:**JAVA PROGRAMMING**

```
import java.util.Scanner;
public class JavaProgrammingLab {
    public static void main(String[] args){
        Scanner in = new Scanner(System.in);
        Cal c = new Cal();
        System.out.println("Enter values");
        int a, b, result = 5, mul;
        a = in.nextInt();
        b = in.nextInt(); c.s
        um(a,
        b); mul = a * b;
        System.out.println("The Sum Is " + result + " Multiplication Is " + mul);
    }
}
```

```
public class Cal {
    void sum(int a, int b) { int
        res = a + b;
        System.out.println(res);
    }
}
```

OUTPUT:

Enter values

3

2

5

The Sum Is 5 Multiplication Is 6

Process finished with exit code 0

MUHAMMAD SALMAN